COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010 Data Structures and Algorithms	**SECTION:**	CPE21S2
DATE PERFORMED:	07/14/2026	**DATE SUBMITTED:**	07/15/2026
NAME:	Jiro Luis A. Katindig	**INSTRUCTOR:**	Engr. Jimlord M. Quejado

ACTIVITY NO. 1 **REVIEW OF C++ PROGRAMMING**

1. PROCEDURE OUTPUT

What procedures did you perform?

ILO A: Create Code That Follows the Basic C++ Code Structure

- Create a function that will take two numbers and display the sum.

```cpp
#include <iostream>

int sumoftwonum( int fNum, int sNum){
    int sum = fNum + sNum;
    return sum;
}

int main() {
    int num1, num2;
    std::cout<<"Enter two numbers: ";
    std::cin >> num1 >> num2;
    std::cout << "The sum of the two numbers is: " << sumoftwonum(num1, num2) << std::endl;
    return 0;
}

```

```
Enter two numbers: 1 5
The sum of the two numbers is: 6
```

Sections	Answer
Header File Declaration Section	`#include <iostream>`
Global Declaration Section	`int sumoftwonum( int fNum, int sNum){`
Class Declaration and Method Definition Section	N/A
Main Function	`int main() { int num1, num2; std::cout<<"Enter two numbers: "; std::cin >> num1 >> num2; std::cout << "The sum of the two numbers is: " << sumoftwonum(num1, num2) << std::endl; return 0; }`
Method Definition	`int sumoftwonum( int fNum, int sNum){ int sum = fNum + sNum; return sum; }`

- Create a function that will return whether variable A is greater than variable B.

```cpp
#include <iostream>

bool greaterVariable(int varA, int varB);

int main() {
    std::cout << "Variable is A greater than variable B: " << greaterVariable(6, 7) << std::endl;
    std::cout << "Variable is A greater than variable B: " << greaterVariable(10, 9) << std::endl;

    return 0;
}


bool greaterVariable(int varA, int varB) {
    if (varA > varB) {
        return true;
    } else {
        return false;
    }
}
```

```
Variable is A greater than variable B: 0
Variable is A greater than variable B: 1
```

Sections	Answer
Header File Declaration Section	`#include <iostream>`
Global Declaration Section	`bool greaterVariable(int varA, int varB);`
Class Declaration and Method Definition Section	N/A
Main Function	`int main() { std::cout << "Variable is A greater than variable B: " << greaterVariable(6, 7) << std::endl; std::cout << "Variable is A greater than variable B: " << greaterVariable(10, 9) << std::endl; return 0; }`
Method Definition	`bool greaterVariable(int varA, int varB) { if (varA > varB) { return true; } else { return false; } }`

- Create a function that will take two Boolean values and display the result of all logical operations then return true if it was a success

```cpp
#include <iostream>

int SumofTwoNum(int fNum, int sNum);
bool isGreaterThan(float num1, float num2);
bool BooleanValues(bool A, bool B);

int SumofTwoNum(int fNum, int sNum) {
    return fNum + sNum;
}

bool isGreaterThan(float num1, float num2){
    return num1 > num2;
}

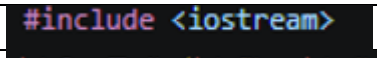
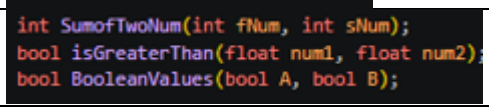
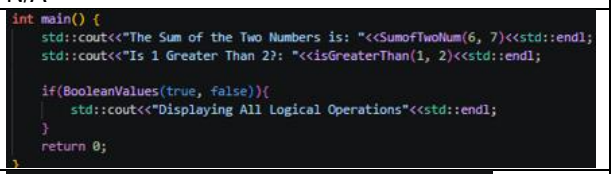
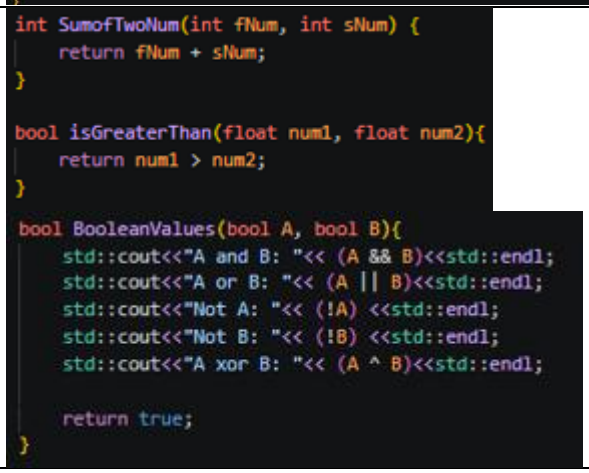
bool BooleanValues(bool A, bool B){
    std::cout<<"A and B: "<< (A && B)<<std::endl;
    std::cout<<"A or B: "<< (A || B)<<std::endl;
    std::cout<<"Not A: "<< (!A) <<std::endl;
    std::cout<<"Not B: "<< (!B) <<std::endl;
    std::cout<<"A xor B: "<< (A ^ B)<<std::endl;

    return true;
}

int main() {
    std::cout<<"The Sum of the Two Numbers is: "<<SumofTwoNum(6, 7)<<std::endl;
    std::cout<<"Is 1 Greater Than 2?: "<<isGreaterThan(1, 2)<<std::endl;

    if(BooleanValues(true, false)){
        std::cout<<"Displaying All Logical Operations"<<std::endl;
    }
    return 0;
}
```

```
The Sum of the Two Numbers is: 13
Is 1 Greater Than 2?: 0
A and B: 0
A or B: 1
Not A: 0
Not B: 1
A xor B: 1
Displaying All Logical Operations
```

Sections	Answer
Header File Declaration Section	
Global Declaration Section	
Class Declaration and Method Definition Section	N/A
Main Function	
Method Definition	

```
#include <iostream>
```

```
int SumofTwoNum(int fNum, int sNum);
bool isGreaterThan(float num1, float num2);
bool BooleanValues(bool A, bool B);
```

```
int main() {
    std::cout<<"The Sum of the Two Numbers is: "<<SumofTwoNum(6, 7)<<std::endl;
    std::cout<<"Is 1 Greater Than 2?: "<<isGreaterThan(1, 2)<<std::endl;

    if(BooleanValues(true, false)){
        std::cout<<"Displaying All Logical Operations"<<std::endl;
    }
    return 0;
}
```

```
int SumofTwoNum(int fNum, int sNum) {
    return fNum + sNum;
}

bool isGreaterThan(float num1, float num2){
    return num1 > num2;
}

bool BooleanValues(bool A, bool B){
    std::cout<<"A and B: "<< (A && B)<<std::endl;
    std::cout<<"A or B: "<< (A || B)<<std::endl;
    std::cout<<"Not A: "<< (!A) <<std::endl;
    std::cout<<"Not B: "<< (!B) <<std::endl;
    std::cout<<"A xor B: "<< (A ^ B)<<std::endl;

    return true;
}
```

ILO B: Implement Appropriate Class Definition and Instances Based on Given Requirements

```cpp
#include <iostream>

class Triangle {
    private:
        double totalAngle, angleA, angleB, angleC;
    public:
        Triangle(double A, double B, double C);
        void setAngles(double A, double B, double C);
        const bool validateTriangle();
};

Triangle::Triangle(double A, double B, double C){
    angleA = A; angleB = B; angleC = C;
    totalAngle = A+B+C;
}

void Triangle::setAngles(double A, double B, double C){
    angleA = A; angleB = B; angleC = C;
    totalAngle = A+B+C;
}

const bool Triangle::validateTriangle() {
    return (totalAngle <= 180);
}

int main(){

    Triangle set1(70, 90, 22);
    std::cout << "The angles are: " << 70 << ", " << 90 << ", " << 22 << std::endl;
    std::cout << "The total angles is: " << 70+90+22 << std::endl;

        if(set1.validateTriangle()){
            std::cout << "The total angles is considered as a Triangle.\n";
        } else {
            std::cout << "The total angles is not considered as a Triangle\n";
        }
        set1.setAngles(50, 90, 27);
        std::cout << "The angles are: " << 50 << ", " << 90 << ", " << 27 << std::endl;
        std::cout << "The total angles is: " << 50+90+27 << std::endl;

        if(set1.validateTriangle()){
            std::cout << "The total angles is considered as a Triangle.\n";
        } else {
            std::cout << "The total angles is not considered as a Triangle\n";
        }

    return 0;
}
```

```
The angles are: 70, 90, 22
The total angles is: 182
The total angles is not considered as a Triangle
The angles are: 50, 90, 27
The total angles is: 167
The total angles is considered as a Triangle.
```

Observation and Comments:

I've observed that my program used class, constructor, and member functions to store and manage the tirangle's angles. At first, I really struggled to understand how these things work, but after testing and debugging I've gotten a hang of it. I was able to make the program work and calculate the angles, and make the validation condition work and check if it produce accurate results.

This activity helped me to understand and recall more about C++ Programming, especially how classes and objects work together. I also learned how important it is to write correct conditinos in a program because even a small mistake can affect the whole output of the code. This improved my debugging and coding skills.

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ILO C: Solve Different Problems using the C++ Programming Language

1. Create a C++ program to swap the two numbers in different variables.
2. Create a C++ program that has a function to convert temperature in Kelvin to Fahrenheit.
3. Create a C++ program that has a function that will calculate the distance between two points.
4. Modify the code given in ILO B and add the following functions:
    a) A function to compute for the area of a triangle
    b) A function to compute for the perimeter of a triangle
    c) A function that determines whether the triangle is acute-angled, obtuse-angled or 'others.'

```cpp
#include <iostream>
#include <cmath>

// TASK 1
void swap(int &a, int &b) {
    int temp = a;
    a = b;
    b = temp;
}

void task1() {

    int num1;
    int num2;

    std::cout << "--TASK 1: Swap Numbers--\n";
    std::cout << "Enter value for num1: ";
    std::cin >> num1;
    std::cout << "Enter value for num2: ";
    std::cin >> num2;

    std::cout << "Before swap: num1 = " << num1 << ", num2 = " << num2 << "\n";

    swap(num1, num2);

    std::cout << "After swap: num1 = " << num1 << ", num2 = " << num2 << "\n";

}

// TASK 2
void task2() {
    double kelvin;
    std::cout << "--TASK 2: Kelvin to Fahrenheit--\n";
    std::cout << "Enter temperature in Kelvin: ";
    std::cin >> kelvin;

    double fahrenheit = (kelvin - 273.15) * 9.0/5.0 + 32.0;
    std::cout << kelvin << " K is equal to " << fahrenheit << " °F\n";
}

// TASK 3
void task3() {
    double x1, y1, x2, y2;
    std::cout << "--TASK 3: Distance between two points--\n";
    std::cout << "Enter coordinates of point 1 (x1 y1): ";
    std::cin >> x1 >> y1;
    std::cout << "Enter coordinates of point 2 (x2 y2): ";
    std::cin >> x2 >> y2;

    double distance = sqrt((x2 - x1) * (x2 - x1) + (y2 - y1) * (y2 - y1));
    std::cout << "Distance between points: " << distance << "\n";
}

// TASK 4
void task4() {
    double a, b, c;
    std::cout << "--TASK 4: Area of a Triangle--\n";
    std::cout << "Enter lengths of the three sides of the triangle (a b c): ";
    std::cin >> a >> b >> c;

    double s = (a + b + c) / 2.0; // semi-perimeter
    double area = sqrt(s * (s - a) * (s - b) * (s - c));
    std::cout << "Area of the triangle: " << area << "\n";
}

// TASK 5
void task5() {
    double a, b, c;
    std::cout << "--TASK 5: Perimeter of a Triangle--\n";
    std::cout << "Enter lengths of the three sides of the triangle (a b c): ";
    std::cin >> a >> b >> c;

    double perimeter = a + b + c;
    std::cout << "Perimeter of the triangle: " << perimeter << "\n";
}

```

```cpp
// TASK 6
void task6() {
    double a, b, c;
    std::cout << "--TASK 6: Triangle Type--\n";
    std::cout << "Enter lengths of the three sides of the triangle (a b c): ";
    std::cin >> a >> b >> c;

    // Check if it's a valid triangle
    if (a + b <= c || a + c <= b || b + c <= a) {
        std::cout << "Invalid triangle!\n";
        return;
    }

    // Determine the type of triangle
    double a2 = a * a;
    double b2 = b * b;
    double c2 = c * c;

    if (a2 == b2 + c2 || b2 == a2 + c2 || c2 == a2 + b2) {
        std::cout << "The triangle is right-angled.\n";
    } else if (a2 > b2 + c2 || b2 > a2 + c2 || c2 > a2 + b2) {
        std::cout << "The triangle is obtuse-angled.\n";
    } else {
        std::cout << "The triangle is acute-angled.\n";
    }
}

int main() {
    int choice;
    std::cout << "Select a program to run:\n";
    std::cout << "1 - Swap Numbers\n";
    std::cout << "2 - Kelvin to Fahrenheit\n";
    std::cout << "3 - Distance between two points\n";
    std::cout << "4 - Area of a Triangle\n";
    std::cout << "5 - Perimeter of a Triangle\n";
    std::cout << "6 - Classify Triangle\n";
    std::cout << "Enter choice (1-6): ";
    std::cin >> choice;

    switch(choice) {
        case 1:
            task1();
            break;
        case 2:
            task2();
            break;
        case 3:
            task3();
            break;
        case 4:
            task4();
            break;
        case 5:
            task5();
            break;
        case 6:
            task6();
            break;
        default:
            std::cout << "Invalid choice!\n";
    }
    return 0;
}
```

```
Select a program to run:
1 - Swap Numbers
2 - Kelvin to Fahrenheit
3 - Distance between two points
4 - Area of a Triangle
5 - Perimeter of a Triangle
6 - Classify Triangle
Enter choice (1-6): 1
--TASK 1: Swap Numbers--
Enter value for num1: 5
Enter value for num2: 4
Before swap: num1 = 5, num2 = 4
After swap: num1 = 4, num2 = 5
--TASK 2: Kelvin to Fahrenheit--
Enter temperature in Kelvin: 278
278 K is equal to 40.73 F
--TASK 3: Distance between two points--
Enter coordinates of point 1 (x1 y1): 10 13
Enter coordinates of point 2 (x2 y2): 11 14
Distance between points: 1.41421
--TASK 4: Area of a Triangle--
Enter lengths of the three sides of the triangle (a b c): 100 50 60
Area of the triangle: 1139.9
--TASK 5: Perimeter of a Triangle--
Enter lengths of the three sides of the triangle (a b c): 100 50 90
Perimeter of the triangle: 240
--TASK 6: Triangle Type--
Enter lengths of the three sides of the triangle (a b c): 90 40 120
The triangle is obtuse-angled.
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

1. Summary of Lessons Learned

In this activity, I learned and recalled how to use and create functions in C++ to solve different programming problems. I practiced assigning values to functions, using mathematical formulas, and organizing my code into separate tasks. I also learned how to use a switch statement to let the user choose which program to run, it's also more convenient for me so that I can code everything on one CPP file instead of making other file for each. Overall, this activity improved my understanding of functions, problem-solving, and writing more organized C++ programs.

2. Analysis of the Procedure

The procedure was easy to follow because each task focused on one programming concept at a time. I first created functions for swapping numbers, converting Kelvin to Fahrenheit, and calculating the distance between two points. After that, I added more functions to the previous triangle program to compute its area, perimeter, and classify its type. I encountered some difficulties while applying the formulas and checking the correct conditions, but by testing and debugging my code and double checking the output to other calculators online, I was able to produce the expected output.

3. Analysis of the Supplementary Activity

The supplementary activity gave me more practice in applying the concepts I learned in class, also from my first year days. It helped me understand how functions can be reused for different tasks and how mathematical formulas can be implemented in C++. Completing all six tasks improved my confidence in coding and strengthened my logical thinking and debugging skills.

4. Concluding Statement / Feedback

I think I did well in this activity because I was able to complete all the required tasks and produce the expected outputs. Although I experienced some challenges while writing and testing the code, I was able to fix my mistakes through debugging. Overall, this activity helped me become more familiar with C++ programming, and I feel more confident in using functions and solving programming problems. I had alot of fun in doing this activity and it motivated me more to learn and review more in C++ programming.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Gemini AI**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Search how to do a code and how to make it work**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **I searched the difference of bool, void, and int so that I have a more clear idea on how to use it. I also searched other codes which I used only as a suggestion or a basis on how will I do it.**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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